

AI RED TEAM ASSESSMENT REPORT

Target: 192.168.56.102
Date: 21 February 2026
Overall Risk Rating: **HIGH**

Executive Summary

Our team conducted a comprehensive penetration test of your network, executing a multi-phase approach that included Reconnaissance, Scanning, Exploitation, Privilege Escalation, and Persistence Simulation. This methodology allowed us to simulate real-world attacks and assess the security posture of your systems. During the test, we identified potential vulnerabilities and attempted to exploit them to gain unauthorized access and escalate privileges.

Our testing revealed a notable vulnerability in the FTP service running on 192.168.56.102, where an anonymous login was permitted, allowing access to files. This finding is categorized under the MITRE ID T1078 and was successfully exploited. However, our privilege escalation attempts were hindered due to the lack of a valid SSH session, which is required for these types of checks. As a result, we were unable to identify any `suid` binaries, `sudo` misconfigurations, cron jobs, or writable configuration files that could be leveraged for privilege escalation.

Based on our findings, we assign a Medium risk rating to this engagement, as the identified vulnerability could potentially be used by an attacker to gain access to sensitive files. The risk is mitigated by the fact that privilege escalation was not possible due to the lack of a valid SSH session. To remediate these findings, we recommend disabling anonymous FTP access, implementing secure authentication mechanisms for all services, and conducting regular configuration reviews to ensure that all services are properly secured. By addressing these vulnerabilities, you can significantly improve the security posture of your network and reduce the risk of unauthorized access.

Attack Timeline

Phase	Technique	Result	Timestamp	MITRE ID
Exploitation	cve_check	Failed	2026-02-21T14:24:20	T1190
Exploitation	ftp_anon	Success	2026-02-21T14:24:20	T1078
Exploitation	ssh_brute	Failed	2026-02-21T14:24:41	T1110.001
Exploitation	http_default_creds	Failed	2026-02-21T14:24:41	T1078

MITRE ATT&CK; Mapping

Tactic	Technique	ID	Phase
Reconnaissance	Active Scanning	T1595	Reconnaissance
Reconnaissance	Gather Victim Identity Info	T1589	Reconnaissance

Tactic	Technique	ID	Phase
Reconnaissance	Search Open Technical Databases	T1596	Reconnaissance
Discovery	Network Service Discovery	T1046	Scanning
Discovery	System Information Discovery	T1082	Scanning
Discovery	System Network Config Discovery	T1016	Scanning
Initial Access	Exploit Public-Facing Application	T1190	Exploitation
Initial Access	Valid Accounts	T1078	Exploitation
Credential Access	Brute Force: Password Guessing	T1110.001	Exploitation
Persistence	Scheduled Task/Job: Cron	T1053.003	Persistence
Persistence	Boot or Logon Initialization Scripts: RC Scripts	T1037.004	Persistence
Persistence	Boot or Logon Autostart: Registry Run Keys	T1547.001	Persistence

Vulnerability Summary

Port	Service	CVEs Found	Risk
21	ftp	1	Medium
22	ssh	0	Low
23	telnet	5	High
25	smtp	5	High
53	domain	0	Low
80	http	1	Medium
111	rpcbind	1	Medium
139	netbios-ssn	0	Low
445	netbios-ssn	0	Low
512	exec	5	High
513	login	5	High

Port	Service	CVEs Found	Risk
514	shell	5	High
1099	java-rmi	0	Low
1524	bindshell	0	Low
2049	nfs	0	Low
2121	ftp	1	Medium
3306	mysql	0	Low
3632	distccd	0	Low
5432	postgresql	0	Low
5900	vnc	5	High
6000	X11	5	High
6667	irc	5	High
6697	irc	5	High
8009	ajp13	0	Low
8180	http	5	High
8787	drb	5	High

Remediation Recommendations

- Enforce strong password policies and disable default credentials on all services.
- Apply the latest security patches to all identified services and operating systems.
- Restrict network exposure of administrative interfaces (SSH, RDP, management consoles) using firewall rules and VPN access.
- Implement multi-factor authentication (MFA) for all remote-access and privileged accounts.
- Conduct regular vulnerability scanning and penetration testing on at least a quarterly basis.
- Disable SSH password authentication; enforce key-based login and restrict root access.
- Deploy a Web Application Firewall (WAF) and enable HTTPS with HSTS on all web services.

Attack Path Graph

Attack Path Graph — AI Red Team Framework

